# Day 06: Transforming SLR models

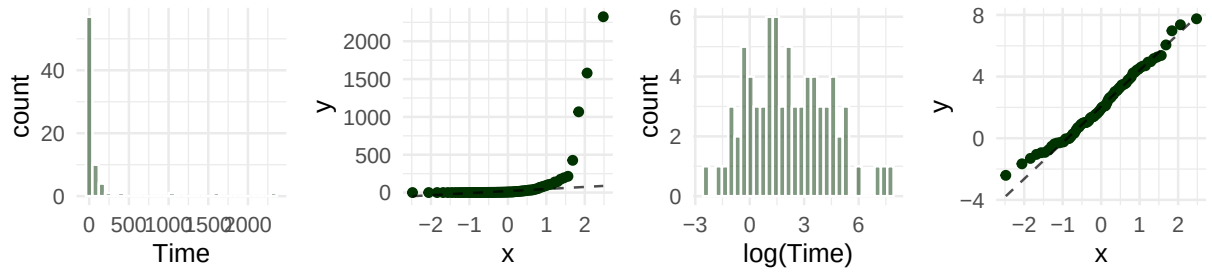
## Last time

We checked: are any conditions for SLR *violated*? Is there a strong predictive relationship between X and Y? **Today:** how can we fix the worst violation?

If the linearity condition is not met, we get systematically biased predictions, might draw false conclusions about the existence or strength of a relationship, and all inference about $\beta_0, \beta_1, \hat{\mu}(Y \mid X)$, Pred$(Y \mid X)$ becomes meaningless.

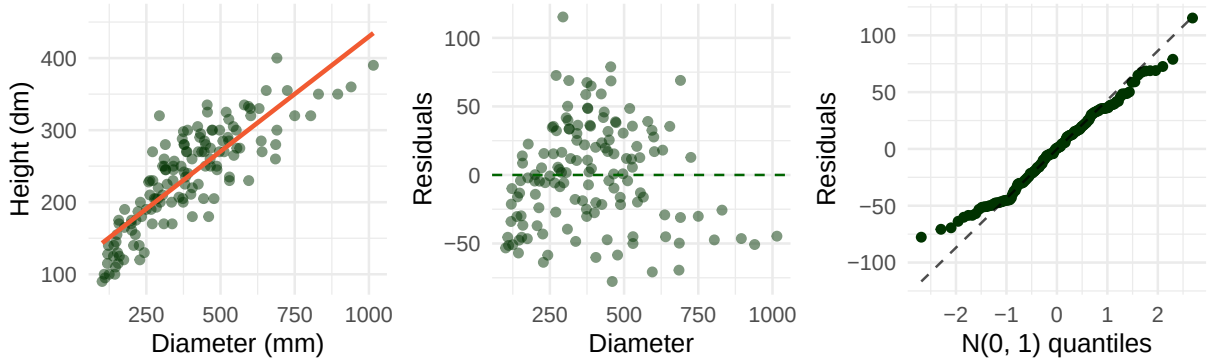
## Easing skew

If a set of data values is skewed to the right, taking the (natural) log of each value *can* result in a data set that is roughly symmetric and often roughly normal.




**New Data for today**: Red Cedar

How are tree height and tree diameter related for the western red cedar?

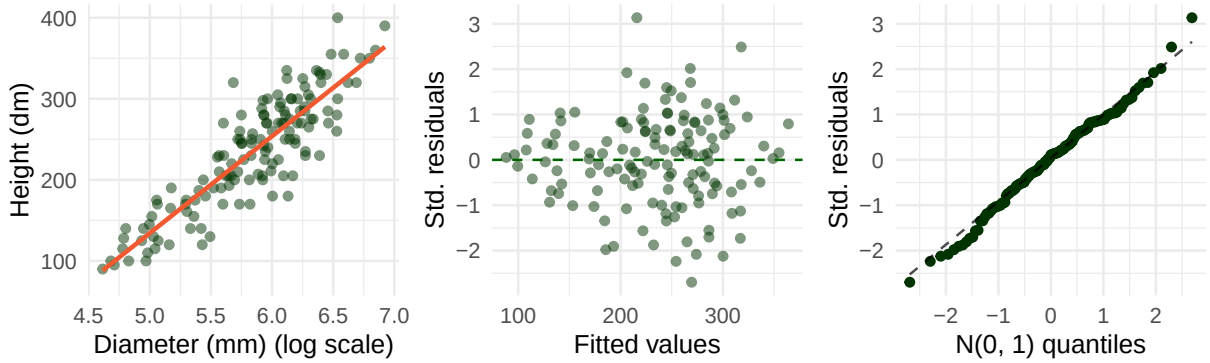


Which condition(s) look violated here: (a) linearity (b) independent errors (c) normal errors (d) equal variance (e) outliers (f) none? **Answer:** _____

Adding `gf_smooth()` to the residual plot can help reveal curvature, but trust your eyes first, and don't over-interpret small samples.

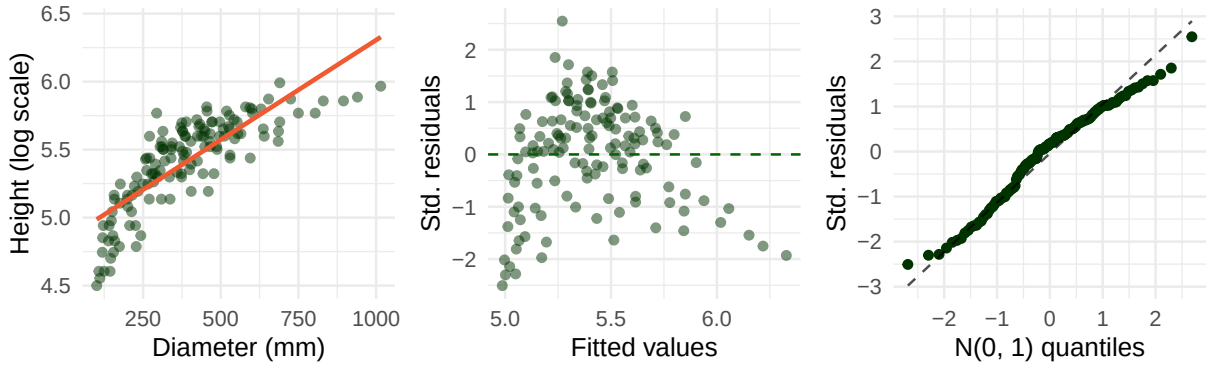
**Does transforming X help?**

Answer:



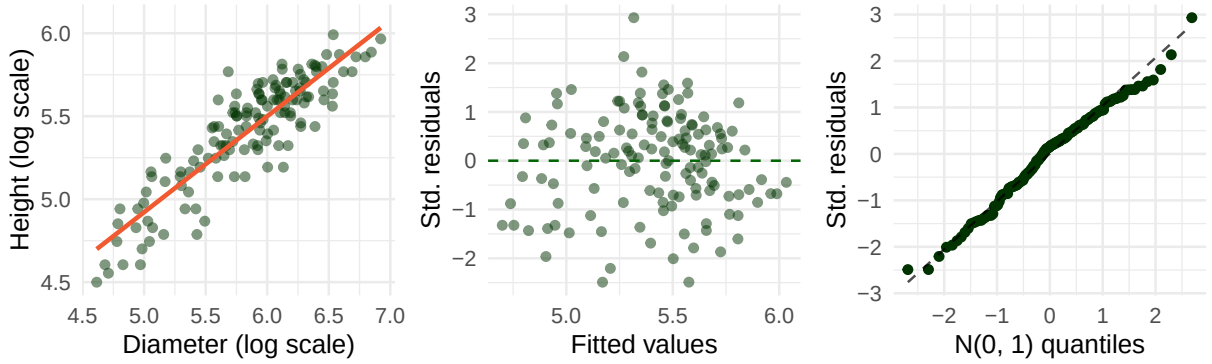

**Does transforming Y help?**

Answer:




**Transforming both X and Y?**

Answer:



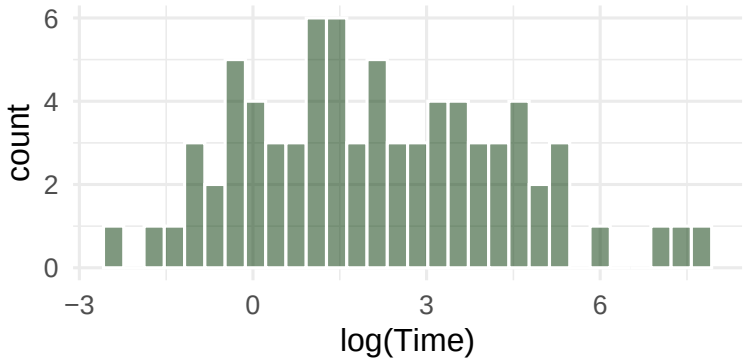

**Group task**

Work through the first example on the handout in your groups (online version with R chunks on Moodle). Compare the three transformation options above – which one best resolves the condition violation(s) you flagged earlier?

## Back-transforming

Converting a transformed variable back to its original scale.

### Example: back-transforming a mean




Mean of `log(Time)` = 2.146.

Back-transformed mean: $e^{2.146} \approx 8.55$.

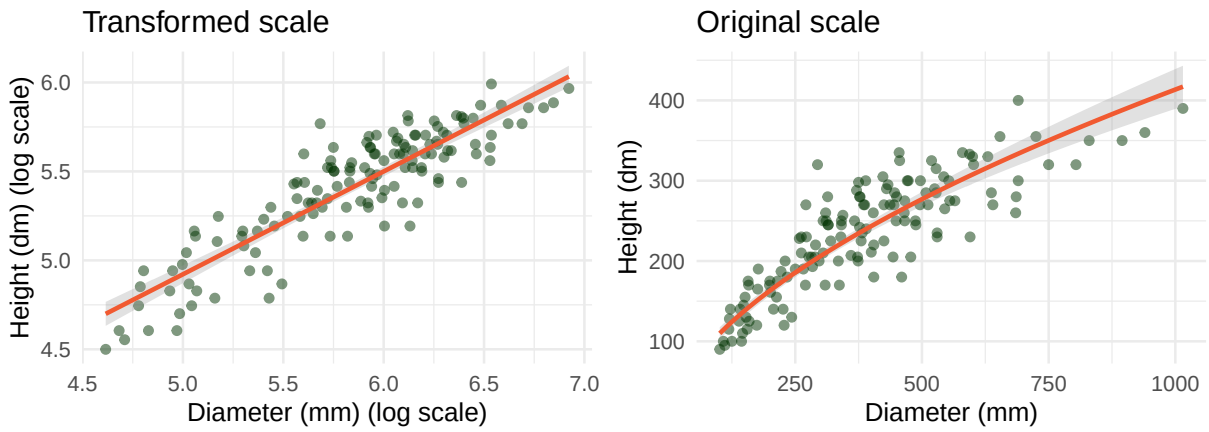
Original `mean(Time)` = 98.558

**Note**

`log()` is the *natural* log; `exp(x)` calculates $e^x$. **Note:** back-transforming the mean of the logs does **not** generally equal the mean on the original scale, but estimates the median instead.

## Displaying a transformed model

`gf_lm(formula = log(y) ~ log(x), backtrans = exp, interval = "confidence")` plots the fitted line and CI back-transformed onto the *original* scale, so you can show results in units your audience understands.




## Rules of thumb

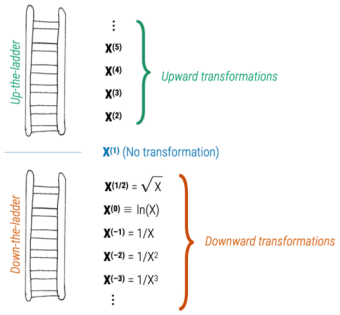
- **Transform X:** if the mean function is nonlinear, but monotonic, and the residual variance is constant.
- **Transform Y:** if the mean function is nonlinear *and* residual variance increases as the mean increases (log, reciprocal, or square root often work).
- **Log rule:** if values range over more than 1 order of magnitude and are strictly positive, the natural log is likely helpful.
- **Range rule:** if the range is considerably less than 1 order of magnitude, transformations are unlikely to help.
- **Square roots** are useful for count data.

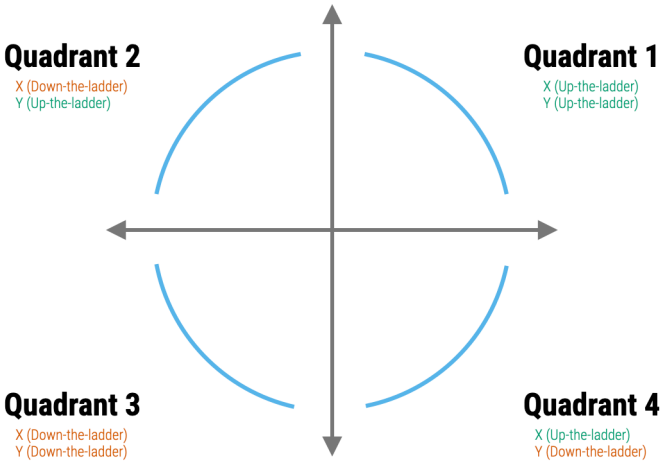
## Ladder of transformations & Rule of the Bulge

Tukey and Mosteller's "Rule of the Bulge" gives a visual guide: match the direction your scatterplot bulges to the ladder position, and move X and/or Y toward that end to straighten the relationship.

## Issues with transformations

- You're often guessing – statistics is an art *and* a science.
- Transformations change the _____ of the parameters – you need to back-transform to give interpretable results.
- Transformations change the SEs of the parameters.
- Not always easy to keep track of all your assumptions once you've transformed.


Up-the-ladder
Down-the-ladder
X(5)
X(4)
X(3)
X(2)
Upward transformations
X(1) (No transformation)
X(1/2) = √X
X(0) ≡ ln(X)
X(−1) = 1/X
X(−2) = 1/X²
X(−3) = 1/X³
Downward transformations


Quadrant 2
X (Down-the-ladder)
Y (Up-the-ladder)
Quadrant 1
X (Up-the-ladder)
Y (Up-the-ladder)
Quadrant 3
X (Down-the-ladder)
Y (Down-the-ladder)
Quadrant 4
X (Up-the-ladder)
Y (Down-the-ladder)